

A Counterexample to Conjecture 1 on Pseudo-Random Array Codes

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Abstract

Conjecture 1 of Blackburn, Chee, Etzion and Lao, *On de Bruijn Array Codes, Part II: Pseudo-Random Array Codes*, asserts a product-closure property for certain pseudo-random array codes under the side condition $n_1 < r_1 < 2n_1$. We give a counterexample with

$$(r_1, r_2; n_1, n_2) = (5, 117; 3, 4).$$

Two explicit irreducible binary polynomials of degree 12 and exponent 585 each yield a $(5, 117; 3, 4)$ -PRAC, while their product does not yield a $(5, 117; 3, 8)$ -PRAC. The failure has a short finite-field certificate: a nonzero binary polynomial supported on the 3 by 8 rectangle vanishes at both evaluation pairs. A separately written recurrence-and-folding implementation exhaustively verifies both hypotheses and exhibits a concrete repeated window. We also prove the conjectured product closure for the subfamily $\text{ord}_{r_1}(2) = n_1$, for every number of factors.

1 Introduction

Etzion developed de Bruijn array codes and their shortened variants in Part I [2]. Blackburn, Chee, Etzion and Lao developed the linear theory in Part II [1]. We use the definitions and folding convention of the version of record of Part II; where its numbering differs from the arXiv preprint, print numbering is used, with the arXiv numbers catalogued in `NUMBERING_MAP.md` of the accompanying repository.

An $(r_1, r_2; n_1, n_2)$ pseudo-random array code, or PRAC, is a collection of binary doubly periodic r_1 by r_2 arrays in which every nonzero binary n_1 by n_2 matrix occurs exactly once as a toroidal window, and which has the shift-and-add property. When $\gcd(r_1, r_2) = 1$, a period- $r_1 r_2$ sequence is folded by placing symbol s_t at the unique coordinate (i, j) satisfying

$$t \equiv i \pmod{r_1}, \quad t \equiv j \pmod{r_2}.$$

Throughout, claims are proved by the argument supplied in the text or verified by the recorded computations that accompany this manuscript, as indicated in context.

The current form of Conjecture 1 in Part II states the following. Suppose

$$n_1 < r_1 < 2n_1$$

and $f_1, \dots, f_\ell$ are irreducible binary polynomials of degree $n_1 n_2$ and exponent $r_1 r_2$, each of whose folded sequences forms an $(r_1, r_2; n_1, n_2)$ -PRAC. Then the conjecture asserts that the product of any k of the factors yields an $(r_1, r_2; n_1, k n_2)$ -PRAC.

The counterexample below satisfies the strict side condition. It therefore resolves Conjecture 1 as stated, but makes no claim about a complete classification of product closure.

2 The evaluation criterion

We first put Theorem 6 of Part II (Theorem 47 in the arXiv version) into a form suited to a short certificate. Let

$$h = n_1, \quad w = n_2, \quad n = hw, \quad e = r_1 r_2.$$

Choose integers μ, ν with $\mu r_1 + \nu r_2 = 1$. If α_u is a root of f_u , define

$$\beta_u = \alpha_u^{\nu r_2}, \quad \gamma_u = \alpha_u^{\mu r_1}.$$

Thus $\alpha_u = \beta_u \gamma_u$, while β_u and γ_u have orders r_1 and r_2 , respectively.

Lemma 1 (Evaluation form of Theorem 6). *For k selected factors, their product folding gives an $(r_1, r_2; h, kw)$ -PRAC if and only if the simultaneous evaluation map*

$$\begin{aligned} \text{ev}: \{P \in \mathbf{F}_2[X, Y] : \deg_X P < h, \deg_Y P < kw\} &\longrightarrow \bigoplus_{u=1}^k \mathbf{F}_{2^n}, \\ P &\longmapsto (P(\beta_u, \gamma_u))_{u=1}^k \end{aligned}$$

is injective.

Proof. For each pair $0 \leq i < h, 0 \leq j < kw$, consider

$$v_{ij} = (\beta_1^i \gamma_1^j, \dots, \beta_k^i \gamma_k^j) \in \bigoplus_{u=1}^k \mathbf{F}_{2^n}.$$

Taking $t_u = \text{Tr}$, as Theorem 6 permits, the u -th block of row (i, j) in the matrix of Theorem 6 consists of the absolute trace pairings of $\beta_u^i \gamma_u^j$ with the basis $1, \alpha_u, \dots, \alpha_u^{n-1}$. The trace pairing is nondegenerate. Consequently the matrix in Theorem 6 has full rank exactly when the kn vectors v_{ij} are linearly independent over $\mathbf{F}_2$. This is equivalent to injectivity of the displayed evaluation map. $\square$

For one factor and a width- w window, Lemma 1 is the linear-independence criterion in Corollary 15 of Part II (arXiv Corollary 48). We use the index range $0 \leq j < w$, as in the proof of that corollary.

3 The counterexample

Theorem 2. *Let*

$$\begin{aligned} f_1(X) &= X^{12} + X^{10} + X^6 + X^5 + X^3 + X + 1, \\ f_2(X) &= X^{12} + X^9 + X^8 + X^4 + X^3 + X + 1. \end{aligned}$$

Both f_1 and f_2 are irreducible of degree 12 and exponent $585 = 5 \cdot 117$. Folding the sequences generated by either factor yields a $(5, 117; 3, 4)$ -PRAC, but folding the sequences generated by $f_1 f_2$ does not yield a $(5, 117; 3, 8)$ -PRAC.

Since $3 < 5 < 6$, Theorem 2 is a counterexample to Conjecture 1.

3.1 Irreducibility and exponent

We encode a binary polynomial by the integer whose bit i is the coefficient of X^i . Thus $f_1 = 0\mathbf{x}146\mathbf{b}$ and $f_2 = 0\mathbf{x}131\mathbf{b}$. Rabin's irreducibility test gives

$$X^{2^{12}} \equiv X \pmod{f_u},$$

and

$$\gcd(f_u, X^{2^6} + X) = \gcd(f_u, X^{2^4} + X) = 1 \quad (u = 1, 2).$$

The exact remainders needed to check the order are

	mod f_1	mod f_2
X^{585}	0x001	0x001
X^{195}	0xab8	0x520
X^{117}	0x4a8	0x7ce
X^{45}	0x19c	0x73d

The first row shows $X^{585} \equiv 1 \pmod{f_u}$, while the remaining rows show $X^{585/p} \not\equiv 1 \pmod{f_u}$ for each prime divisor $p \in \{3, 5, 13\}$ of 585. Together these prove that each residue class of X has exact order 585.

3.2 The two single-factor tests

Work in

$$K = \mathbf{F}_2[a]/(f_1(a)).$$

Direct polynomial reduction shows that $a_2 = a^{46}$ is a root of f_2 . We take $\alpha_1 = a$ and $\alpha_2 = a_2$. Use

$$47 \cdot 5 - 2 \cdot 117 = 1$$

and define

$$b = a^{351} = a_2^{351}, \quad c_1 = a^{235}, \quad c_2 = a_2^{235} = a^{280}.$$

The element b has order 5, so

$$F = \mathbf{F}_2(b) \cong \mathbf{F}_2[B]/(B^4 + B^3 + B^2 + B + 1) = \mathbf{F}_{16}.$$

Both c_1 and c_2 have degree three over F . Their distinct minimal polynomials over F are

$$\begin{aligned} m_1(Y) &= Y^3 + (1 + b + b^2 + b^3)Y^2 + b^2Y + (b^2 + b^3), \\ m_2(Y) &= Y^3 + (1 + b + b^2 + b^3)Y^2 + (1 + b + b^3)Y + (b^2 + b^3). \end{aligned}$$

Put

$$U = \text{span}_{\mathbf{F}_2}\{1, b, b^2\} \subset F.$$

The binary 3 by 4 rectangle evaluates at $X = b$ to the $\mathbf{F}_2$ -space $U[Y]_{<4}$. If a nonzero member of this space vanished at c_u , it would have the form $q m_u$, where

$$q = q_0 + q_1b + q_2b^2 \in U \setminus \{0\}.$$

Requiring the b^3 -coordinate of all four coefficients to vanish gives, for either m_u ,

$$q_0 = 0, \quad q_0 + q_2 = 0.$$

The coefficient of Y then gives $q_1 + q_2 = 0$ for m_1 , or $q_0 + q_1 + q_2 = 0$ for m_2 . Hence $q_0 = q_1 = q_2 = 0$, a contradiction. Lemma 1 therefore proves both single-factor PRAC claims.

3.3 A rectangle relation for the product

Multiplication in $F[Y]$ gives

$$\begin{aligned} M(Y) := m_1(Y)m_2(Y) &= Y^6 + (1 + b + b^2)Y^4 + b^3Y^3 \\ &\quad + (1 + b^2 + b^3)Y^2 + (b + b^2)Y + (1 + b^2 + b^3). \end{aligned}$$

Although M has coefficients outside U , multiplication by $Y + b$ cancels all b^3 -coordinates:

$$\begin{aligned} A(Y) := (Y + b)M(Y) &= Y^7 + bY^6 + (1 + b + b^2)Y^5 + (b + b^2)Y^4 \\ &\quad + bY^3 + (1 + b)Y^2 + Y + (1 + b^2). \end{aligned}$$

Thus $A \in U[Y]_{<8} \setminus \{0\}$, and

$$A(c_1) = A(c_2) = 0.$$

Expressing each coefficient in the basis $1, b, b^2$ produces the binary rectangle polynomial

$$\begin{aligned} P(X, Y) &= 1 + Y + Y^2 + Y^5 + Y^7 \\ &\quad + X(Y^2 + Y^3 + Y^4 + Y^5 + Y^6) \\ &\quad + X^2(1 + Y^4 + Y^5). \end{aligned}$$

It is nonzero, $\deg_X P < 3$, $\deg_Y P < 8$, and

$$P(b, c_1) = P(b, c_2) = 0.$$

Lemma 1 now proves that the product fails.

There is also a basis-free univariate certificate. For a root α_u of either factor,

$$\beta_u = \alpha_u^{351}, \quad \gamma_u = \alpha_u^{235}.$$

Reducing exponents modulo 585, the rectangle relation becomes

$$\begin{aligned} Q(Z) &= P(Z^{351}, Z^{235}) \pmod{Z^{585} - 1} \\ &= 1 + Z^5 + Z^6 + Z^{117} + Z^{121} + Z^{122} + Z^{235} + Z^{236} \\ &\quad + Z^{356} + Z^{470} + Z^{471} + Z^{472} + Z^{475}. \end{aligned}$$

Direct binary polynomial division gives

$$Q(Z) \equiv 0 \pmod{f_1(Z)}, \quad Q(Z) \equiv 0 \pmod{f_2(Z)}.$$

This sparse identity is an independently checkable certificate of the rank failure; it uses neither a field basis nor a trace convention.

4 Exhaustive verification by a separate implementation

A separate implementation generated sequences directly from the binary linear recurrences, folded them using only the CRT rule, and exhaustively enumerated all windows. It was written with knowledge of the claimed witness, but reuses no source code from the algebraic path and shares no finite-field, trace, determinant, or evaluation machinery with the proof above.

For each single factor, the checker enumerated

$$\frac{2^{12} - 1}{585} = 7$$

shift-orbit representatives and scanned all $7 \cdot 585 = 4095$ windows. Every nonzero 3 by 4 binary matrix occurred exactly once, and no zero window occurred.

For the product, the checker enumerated

$$\frac{2^{24} - 1}{585} = 28,679$$

shift-orbit representatives and scanned

$$28,679 \cdot 585 = 16,777,215 = 2^{24} - 1$$

windows. The results were:

quantity	count
distinct nonzero windows observed	8,388,607
missing nonzero windows	8,388,608
duplicate nonzero occurrences	8,388,607
zero-window occurrences	1

With rows written left to right, the window

```
11110001
00010011
11100011
```

occurs in two distinct shift orbits. Using zero-based coordinates, its two frozen occurrences are

representative state	top	left
1	1	40
8	4	28

The first missing nonzero window is

```
00100000
00000000 ,
00000000
```

and an all-zero window occurs in representative state 10781 at (top, left) = (4, 14).

The frozen JSON record has SHA-256

```
ba098b857e2aeecccf726e3440736dd25969732cd6c0bac0cb984a43efaa3b97,
```

and the checker has SHA-256

```
ed84442dc519ff07fc24c27f00d4b91366011c89a24e70562349351875ba23f0.
```

The run used Python 3.12.13 on Linux and no randomness. The complete record and checker are supplied as `implementations/bruteforce/counterexample_frozen.json` and `implementations/bruteforce/bruteforce.py`.

5 Why the side condition was not enough

The counterexample suggests a useful correction to the structural picture. We retain h, w, r_1, r_2, n from Section 2 and write

$$d = \text{ord}_{r_1}(2).$$

Lemma 3 (Normalisation). *If one single factor passes and $h < r_1 < 2h$, then*

$$d = \varphi(r_1), \quad h \leq d < 2h.$$

Consequently, after replacing each selected root by a binary Frobenius conjugate, all its r_1 -components may be chosen equal:

$$\beta_1 = \dots = \beta_k = b.$$

Proof. The single-factor basis contains $1, \beta, \dots, \beta^{h-1}$, so these powers are independent and

$$d = [\mathbf{F}_2(\beta) : \mathbf{F}_2] \geq h.$$

Since $d \mid \varphi(r_1)$ and $\varphi(r_1) \leq r_1 - 1 < 2h$, the positive integer $\varphi(r_1)/d$ is less than two and hence equals one. Thus all primitive r_1 -th roots form one Frobenius orbit, proving the normalisation claim. $\square$

Let

$$F = \mathbf{F}_2(b) = \mathbf{F}_{2^d}, \quad U = \text{span}_{\mathbf{F}_2}\{1, b, \dots, b^{h-1}\}.$$

Let $g_u(Y) \in F[Y]$ be the minimal polynomial of γ_u over F , and put $m = n/d$. Since $\alpha_u = b\gamma_u$ and $K = \mathbf{F}_2(\alpha_u)$, one has $F(\gamma_u) = K$; hence $\deg g_u = m$. Distinct binary factors give distinct g_u : equality would make γ_v an F -Frobenius conjugate of γ_u , and therefore α_v a binary Frobenius conjugate of α_u .

The evaluation criterion now becomes the exact coefficient-space condition

$$U[Y]_{<kw} \cap \left(\prod_{u=1}^k g_u(Y) \right) = \{0\},$$

where the ideal is taken in $F[Y]$. The individual hypotheses are

$$U[Y]_{<w} \cap (g_u) = \{0\}.$$

If $d = h$, then $U = F$, and ordinary degree counting proves product closure. If $d > h$, then U is only a proper additive subspace of F , and $m = hw/d < w$. A low-degree multiple of $\prod g_u$ can therefore have all its coefficients in U . In Theorem 2, $d = 4$, $h = 3$, $m = 3$, $w = 4$, and the multiplier is exactly $Y + b$.

Theorem 4 (A product-closed subfamily). *Let h, w, r, s be positive integers such that*

$$\gcd(r, s) = 1, \quad h < r < 2h, \quad \text{ord}_{rs}(2) = hw, \quad \text{ord}_r(2) = h.$$

For every collection of k distinct irreducible binary polynomials of degree hw and exponent rs , folding the sequences generated by their product yields an $(r, s; h, kw)$ -PRAC.

Proof. Since $\text{ord}_r(2) = h$, while $\text{ord}_r(2) \mid \varphi(r) \leq r - 1 < 2h$, we have $\text{ord}_r(2) = \varphi(r)$. Thus all primitive order- r elements form one binary Frobenius orbit, and we may choose the roots so their order- r components all equal b . Here

$$F = \mathbf{F}_2(b) = \mathbf{F}_{2^h}$$

and $1, b, \dots, b^{h-1}$ is a basis of F , so $U = F$. For each selected root, its order- s component γ_u has degree

$$[\mathbf{F}_{2^{hw}} : F] = w$$

over F . The corresponding minimal polynomials $g_u \in F[Y]$ are distinct, by the argument preceding the theorem. A binary rectangle relation would give a polynomial $A \in F[Y]$ of degree less than kw that vanishes at every γ_u . It would therefore be divisible by the product of the k distinct degree- w polynomials g_u , which has degree kw . Hence $A = 0$, and then all binary rectangle coefficients are zero. Lemma 1 completes the proof. $\square$

Theorem 4 covers the sufficient-condition family in Theorem 5 of Part II (Theorem 30 in the arXiv version) under the conjecture’s side condition. It includes the motivating $(5, 17; 4, 2)$ example and the $(3, 7; 2, 3)$ exponent-21 example.

Lemma 17 of Part II (Lemma 44 in the arXiv version) does not prove this result. That lemma concerns the paper’s $\vee$ -construction, with separate degree- n_1 , exponent- r_1 and degree- n_2 , exponent- r_2 factors. Conjecture 1 multiplies arbitrary degree- $n_1 n_2$, exponent- $r_1 r_2$ factors. In the counterexample there is no degree-three subfield containing b , so the required decomposition is absent.

6 Conclusion

Theorem 2 settles Conjecture 1 in the negative. The sparse polynomial P is a compact certificate: it is a nonzero member of the proposed 3 by 8 window space and vanishes at both selected evaluation pairs. The separately written exhaustive recurrence computation confirms both single-factor hypotheses and the product failure.

Theorem 4 of this note (unrelated to Theorem 4 of the published source paper, the join-construction result) is a sufficient condition for product closure: it covers the exponent-85 motivating family and identifies a broad sufficient regime. Outside that regime, the note gives a coefficient-space intersection criterion, not a complete classification: product closure is equivalent to

$$U[Y]_{<kw} \cap \left(\prod g_u\right) = \{0\}.$$

This formulation may be useful for classifying the remaining parameters.

AI-assistance disclosure

This research was produced by an AI research pipeline directed by Cathal Ryan Hynes at P.I.G.M.I.E. Ltd., Ireland. From a research brief written by Anthropic’s Claude (Fable), OpenAI’s Sol Ultra, running in the ChatGPT web interface, found the counterexample and the sufficient-condition theorem; Fable separately rebuilt the entire verification, reusing no source code, and matched every count; a separate Sol Ultra session contributed two further audit rounds. Cathal Ryan Hynes set the target and the verification standard, checked the certificate himself, and stands behind this release. The exhaustive recurrence checker, the separate algebra implementation, the third finite-field implementation, the audit reimplementations, and the frozen outputs and hashes accompany this

manuscript. No claim is made beyond resolution of Conjecture 1 as stated in the version of record and the explicit sufficient-condition theorem proved here.

References

- [1] S. R. Blackburn, Y. M. Chee, T. Etzion, and H. Lao, On de Bruijn array codes, Part II: Pseudo-random array codes, *IEEE Transactions on Information Theory*, 72(8):6204–6221, August 2026. doi:10.1109/TIT.2026.3686267. Also arXiv:2501.12124v4 (last revised 19 August 2025).
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